

Project #7: Warehouse Automation with an Autonomous Fork-Lift Mobile Robot

Suggested title: WMS-Integrated Autonomous Omnidirectional Forklift for Smart Warehouse Material Handling

Project Type

Continuation and major upgrade of last year's prototype

Core Technologies

SLAM • computer vision • embedded control • IoT/WMS

Main Outcome

Safe autonomous pallet pickup, transport and placement

Existing Autonomous Forklift Prototype and Material-Handling Demonstration



Prototype platform



Pallet / box transport



Front lifting module

Figure 1. Existing autonomous forklift prototype, pallet-transport test and front lifting assembly.

Project Description

This graduation project aims to transform the existing forklift prototype into a **fully autonomous, WMS-connected mobile material-handling robot**. The vehicle will receive transport missions from a Warehouse Management System (WMS), localize itself within the warehouse, plan collision-free routes, identify and align with pallets, lift the assigned load, transport it safely, and confirm task completion. The upgraded system is intended to replace repetitive manual transport while improving traceability, throughput, safety and operational consistency.

The project combines mechanical redesign, embedded control, mobile-robot navigation, computer vision, industrial communication, battery management and autonomous charging. A key design decision will be a comparative assessment of **Mecanum-wheel omnidirectional motion versus an enhanced differential-drive configuration** to achieve reliable traction, torque distribution and maneuverability in narrow aisles and near storage racks.

Why the Project Is Important and Its Main Objectives

Industrial Importance

- Reduces manual handling and repetitive driving.
- Improves material traceability through WMS task logging.
- Enables safer operation around workers and dynamic obstacles.
- Provides a scalable platform for smart factories and Industry 4.0 warehouses.

Primary Project Objectives

- Upgrade the drive, suspension, lift mechanism and battery system.
- Implement SLAM, path planning, obstacle avoidance and pallet-pose estimation.
- Develop WMS/IoT communication, TCP camera streaming and a QML GUI.
- Add low-battery navigation, autonomous docking, charging and mission resumption.

Proposed System Architecture and Technical Work Packages

The architecture separates mission-level autonomy, safety-critical real-time control, perception, communication and energy management.

Autonomous Forklift System Architecture and Information Flow

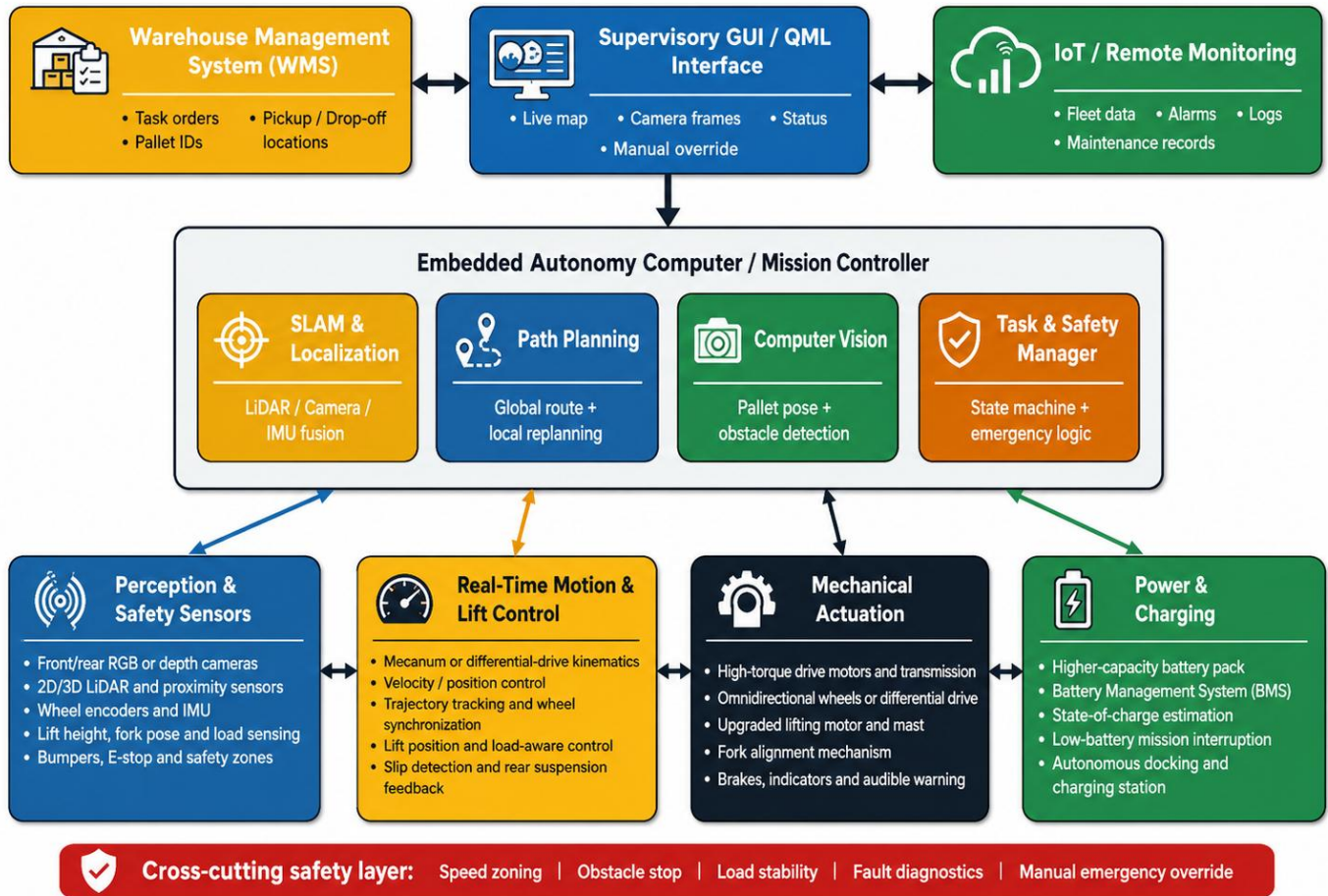


Figure 2. Integrated architecture for WMS communication, autonomous navigation, perception, motion/lift control and charging.

Major Technical Development Areas

Work Package	Development Scope	Expected Engineering Result
1. Mechanical platform	Evaluate differential-drive alternatives; improve rear suspension and traction; strengthen drive motor, transmission and chassis; increase lift-motor torque.	Compact, stable platform capable of narrow-aisle motion and heavier pallet handling.
2. Embedded control	Implement wheel-velocity synchronization, trajectory tracking, lift-position control, load-aware limits, slip detection and emergency-stop logic.	Accurate low-level motion, smooth pallet lifting and deterministic safety behavior.
3. Localization and navigation	Fuse LiDAR/camera/IMU/encoders for SLAM; develop global path planning and local replanning around static and moving obstacles.	Reliable autonomous navigation with repeatable localization accuracy.
4. Vision and pose estimation	Use front and rear cameras for obstacle detection, pallet/fork pose estimation, alignment verification and optional gesture commands.	Robust pallet approach and improved safety during forward and reverse motion.
5. WMS, IoT and GUI	Exchange missions and status with the WMS; stream camera frames through TCP; build a QML interface for map, alarms, task queue and manual override.	Real-time fleet supervision, traceable task execution and remote monitoring.
6. Energy autonomy	Increase battery capacity; design BMS and state-of-charge estimation; develop charging dock detection, alignment and controller.	Automatic low-battery recovery, charging and mission resumption.

Recommended Sensors, Controllers and Interfaces

Sensors: front/rear RGB or depth cameras, 2D/3D LiDAR, wheel encoders, IMU, proximity sensors, bumpers, lift-height sensing, fork-position sensing, motor current/load estimation and battery sensors. **Computing:** embedded Linux computer for perception/SLAM and a real-time microcontroller for motor, lift and safety control. **Interfaces:** CAN/UART/Ethernet internally; Wi-Fi or industrial wireless connection to WMS/IoT services; TCP/QML for live visualization and remote supervision.

Implementation of Workflow, Deliverables and Evaluation

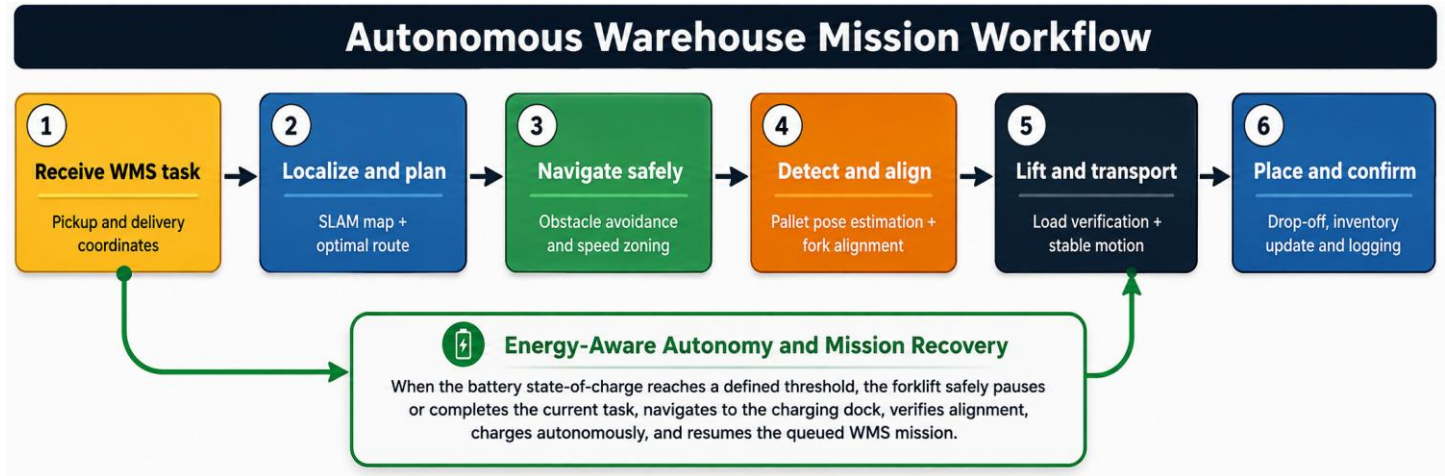


Figure 3. End-to-end autonomous mission sequence with energy-aware docking and task resumption.

Planned Implementation Methodology

Development Phases	Key Verification Activities
1. Audit the existing prototype, identify mechanical/electrical limitations and define load, speed and aisle-width requirements. 2. Upgrade hardware, battery and safety hardware. 3. Build the vehicle model, calibrate sensors and implement low-level drive/lift control. 4. Develop mapping, localization, path planning, obstacle avoidance and pallet alignment. 5. Integrate the WMS/IoT communication layer, TCP camera streaming and QML GUI. 6. Implement autonomous charging, conduct staged tests, and complete full warehouse missions.	- Mechanical tests: payload, lift height, traction, turning radius, slope/bump performance and structural stability. - Control tests: wheel tracking error, lift-position accuracy, stopping distance and emergency response time. - Autonomy tests: mapping repeatability, localization error, route success rate and dynamic-obstacle avoidance. - Vision tests: pallet detection rate, pose/alignment error and rear-camera safety coverage. - System tests: WMS task latency, GUI update rate, mission completion time and communication reliability. - Energy tests: operating duration, BMS accuracy, docking success and charge-to-resume time.

Previous-Years Resources and Contact

Previous-Year Prototype Resources

1. [YouTube demonstration](#)
2. [LinkedIn project post](#)
3. [Google Drive project file](#)

For More details contact

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The student team will build on the existing platform while delivering a safer, more powerful and fully autonomous warehouse solution.